CPPC 18/L

COMPUTER NETWORK & SECURITY

ACTIVITY No.8

DNS

Submitted by:

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BS CpE 3B

Submitted to:

ENGR. JAN ERROL MAMPUSTI

Subject Professor

I. Objectives

- To configure a DNS server for several client computers.
- To connect a domain name to a specific server IP address.
- To understand how DNS translates a website name into an IP address.
- To access a customized webpage using a domain name instead of typing the numerical IP address.
- To verify if DNS and HTTP services are working properly in the network.

II. Materials

- Cisco Packet Tracer
- 1 Server
- 1 Switch
- 2–3 PCs
- Laptop
- Copper Straight-Through cables

III. Procedures

1. Create the Network Topology

- Open Cisco Packet Tracer.
- Add the required devices in the workspace:
  - 1 Server
  - 1 Switch
  - 3 PCs
- Connect all devices using Copper Straight-Through cables.
- Connect the server to the switch.
- Connect each PC to the same switch.
- Make sure that all connections are active and properly linked.

2. Configure the Server IP Address

- Click the Server.
- Go to Desktop.
- Open IP Configuration.
- Set the server with the following network details:
  - IP Address: 192.168.1.1
  - Subnet Mask: 255.255.255.0
  - DNS Server: 192.168.1.1
- The server will act as both the web server and DNS server in the network.

3. Enable the DNS Service

- Click the Server.
- Go to the Services tab.
- Select DNS.
- Turn the DNS service ON.
- This allows the server to translate a domain name into its corresponding IP address.

4. Add a DNS Record

- In the DNS settings, create a new DNS record.
- Enter the following information:
  - Name: www.facebook.com
  - Address: 192.168.1.1
- Click Add to save the DNS record.
- This means that when the domain name is entered in the browser, it will point to the server's IP address.

5. Enable HTTP and Edit the Webpage

- Go to Server.
- Open the Services tab.
- Select HTTP.
- Make sure that the HTTP service is turned ON.
- Open or edit the webpage file, such as:
  - helloworld.html
- Replace the default content with the customized webpage content.
- Save the changes after editing.
- This allows the server to display the customized webpage when accessed by the PCs.

6. Configure the PCs

- Click each PC one by one.
- Go to Desktop.
- Open IP Configuration.
- Set a static IP address for each PC.
- Use the following network configuration:
  - PC0
    - IP Address: 192.168.1.10
    - Subnet Mask: 255.255.255.0
    - DNS Server: 192.168.1.1
  - PC1
    - IP Address: 192.168.1.11
    - Subnet Mask: 255.255.255.0
    - DNS Server: 192.168.1.1

- PC2
    - IP Address: 192.168.1.12
    - Subnet Mask: 255.255.255.0
    - DNS Server: 192.168.1.1

- Make sure all PCs are in the same network as the server.
- The DNS server address should match the server's IP address.

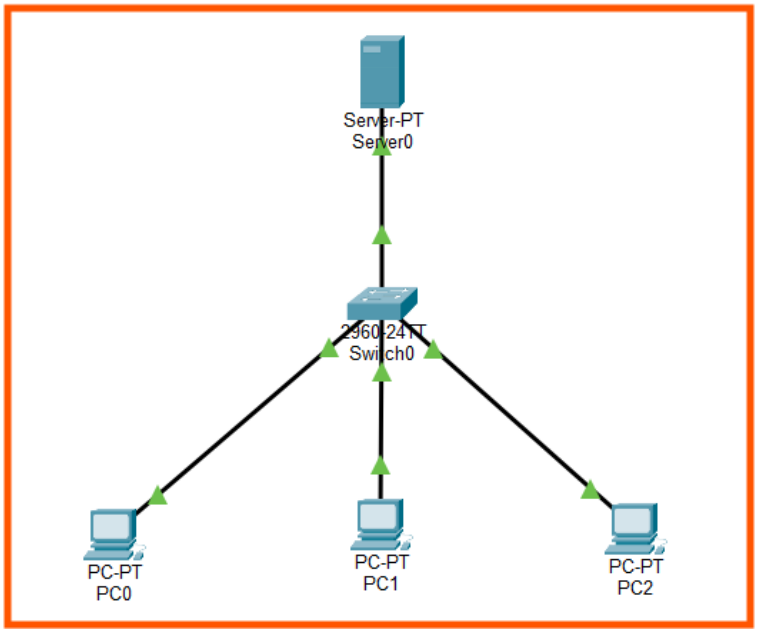
7. Test the DNS and HTTP Services

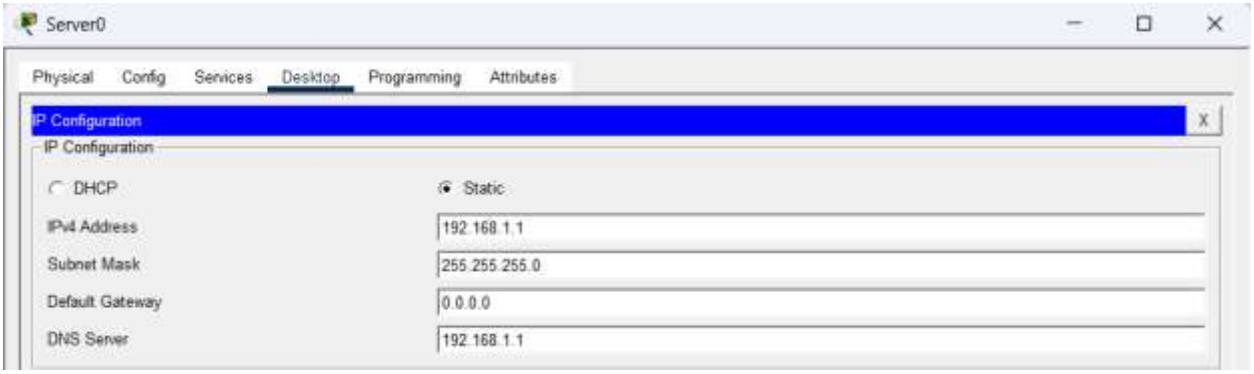
- Click each PC.
- Go to Desktop.
- Open Web Browser.
- In the URL bar, type:
    - www.facebook.com
- Click Go.
- The customized webpage should load if the DNS and HTTP configurations are correct.
- Repeat the same process on the other PCs to confirm that all clients can access the webpage using the domain name.

IV. Conclusion

The activity showed how DNS works by converting a domain name into an IP address within a local network. By configuring the DNS server and linking the domain name **www.facebook.com** to the server's IP address, the PCs were able to access the customized webpage without typing the numerical IP address. This proves that DNS makes network access easier for users, while the successful webpage loading also confirms that both DNS and HTTP services were properly configured and functioning correctly.

V. Documentation





Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DNS
DNS Service
On
Off
Resource Records
Name
www.facebook.com
Type
A Record
Address
192.168.1.1
Add
Save
Remove
No.
Name
Type
Detail
0
intagram
A Record
192.168.1.1
1
tiktok
A Record
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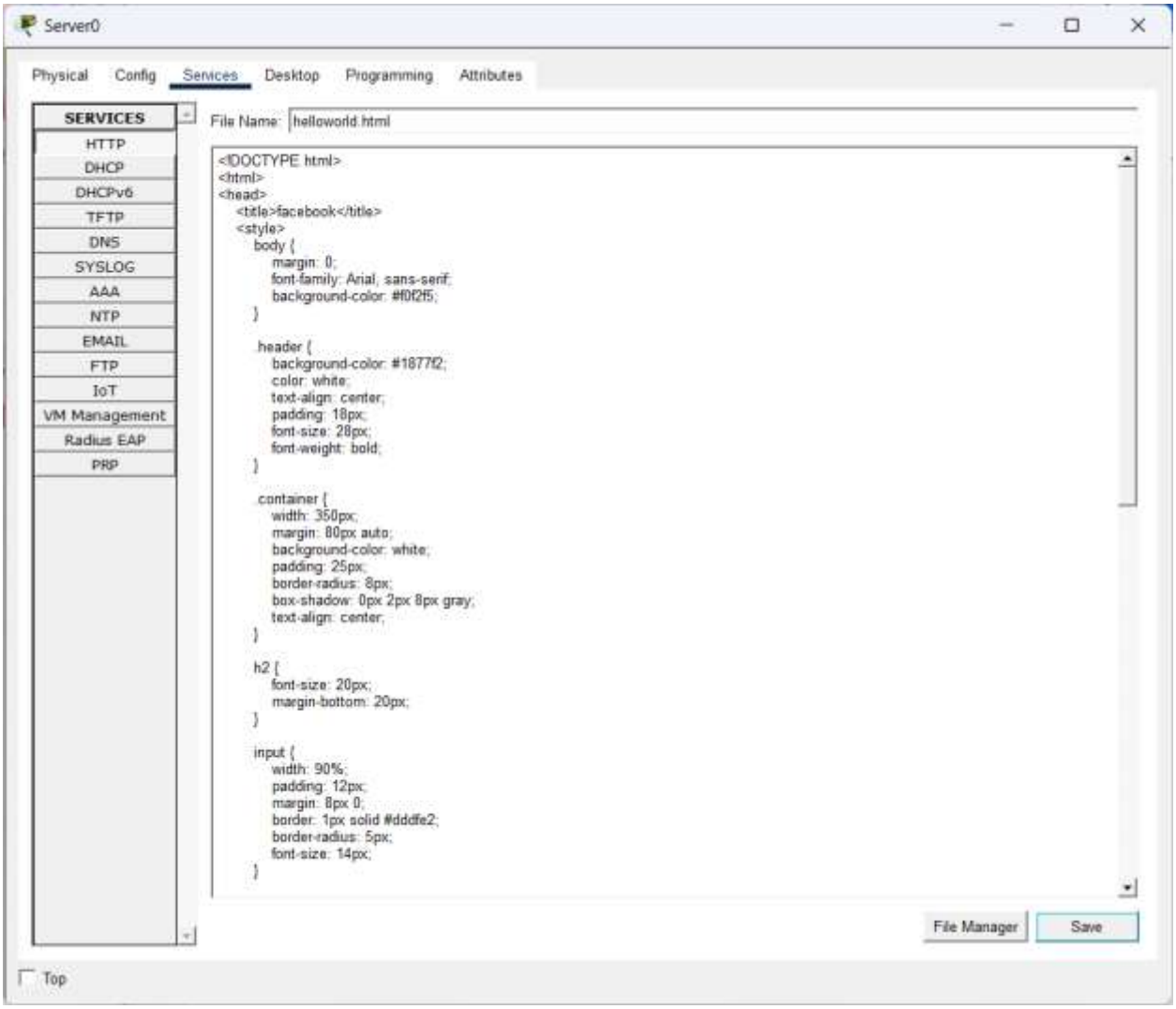
2
www.facebook.com
A Record
192.168.1.1


Server0
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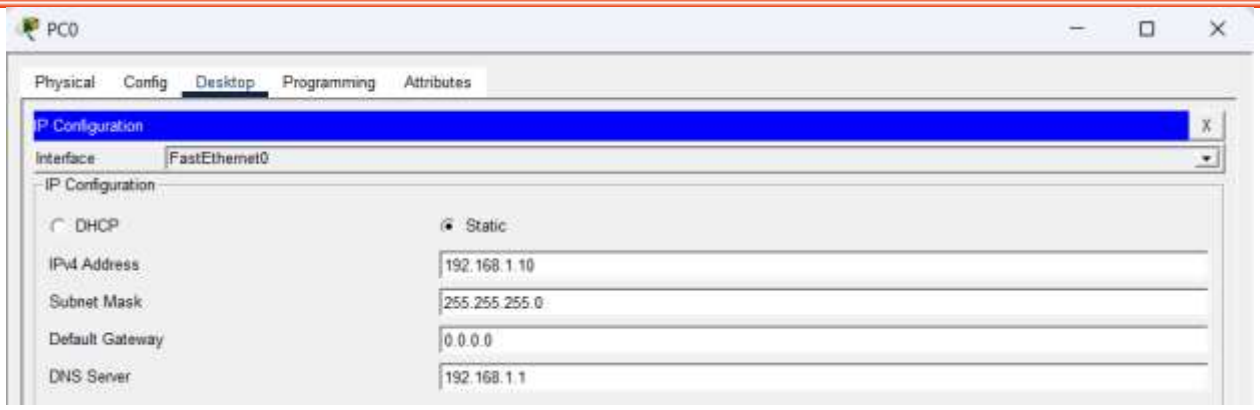
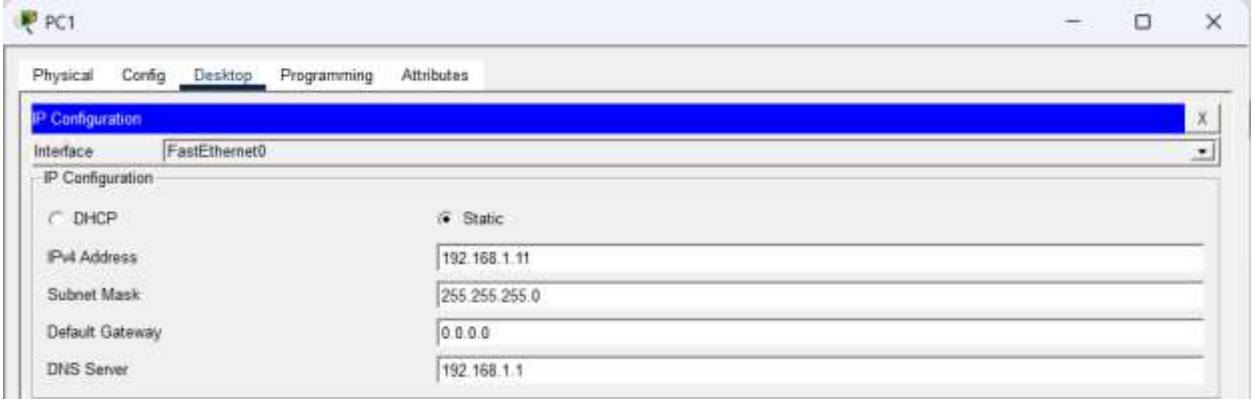
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
File Name:
helloworld.html
<!DOCTYPE html>
<html>
<head>
<title>facebook</title>
<style>
body {
margin: 0;
font-family: Arial, sans-serif;
background-color: #f0f2f5;
}
.header {
background-color: #1877f2;
color: white;
text-align: center;
padding: 18px;
font-size: 28px;
font-weight: bold;
}
.container {
width: 350px;
margin: 80px auto;
background-color: white;
padding: 25px;
border-radius: 8px;
box-shadow: 0px 2px 8px gray;
text-align: center;
}
h2 {
font-size: 20px;
margin-bottom: 20px;
}
input {
width: 90%;
padding: 12px;
margin: 8px 0;
border: 1px solid #dddfe2;
border-radius: 5px;
font-size: 14px;
}
File Manager
Save
Top


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.10
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
192.168.1.1


PC1
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Subnet Mask
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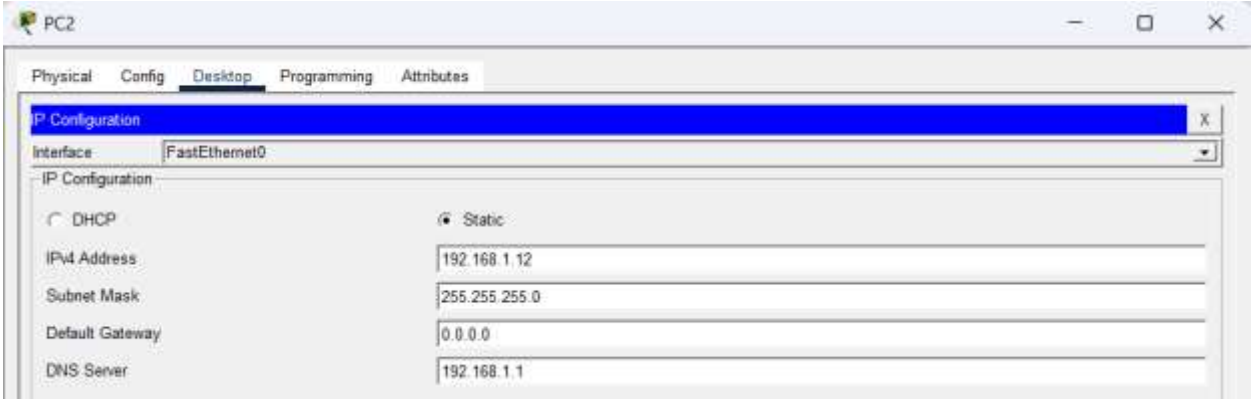
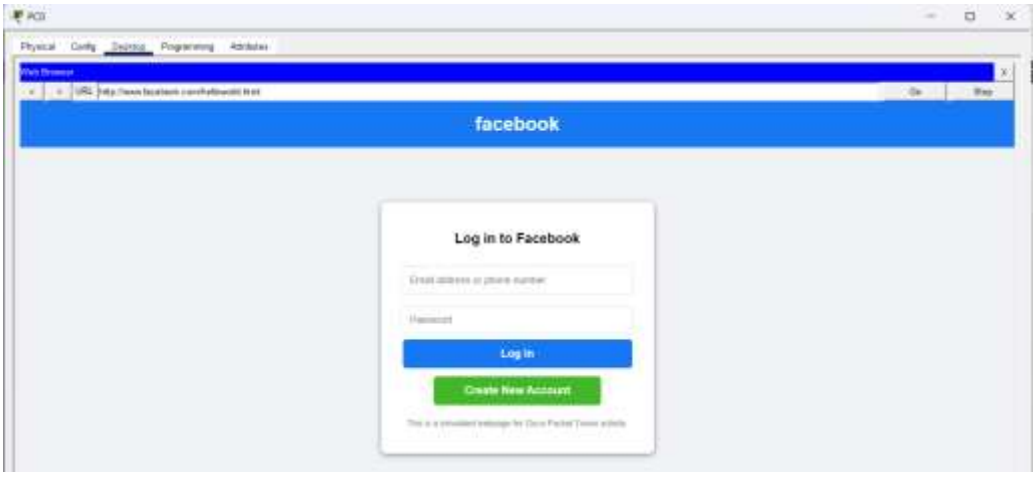
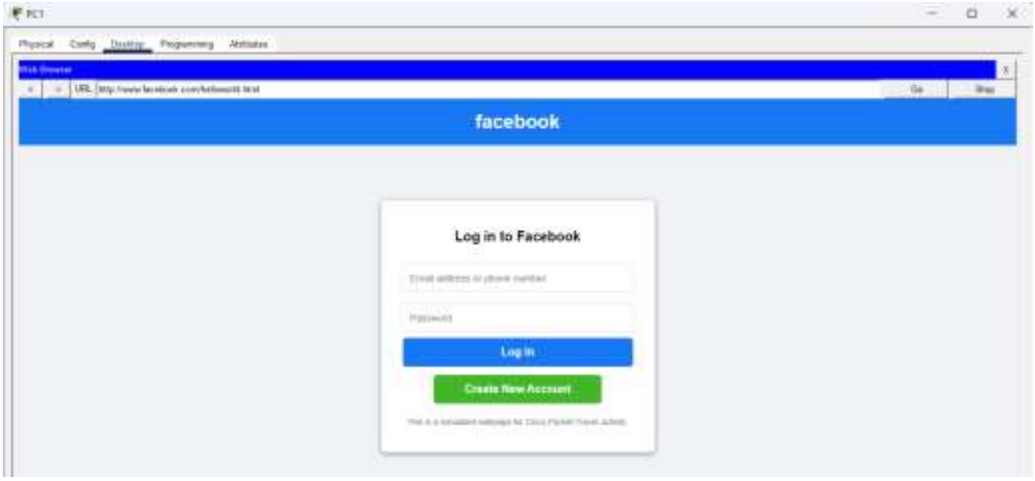
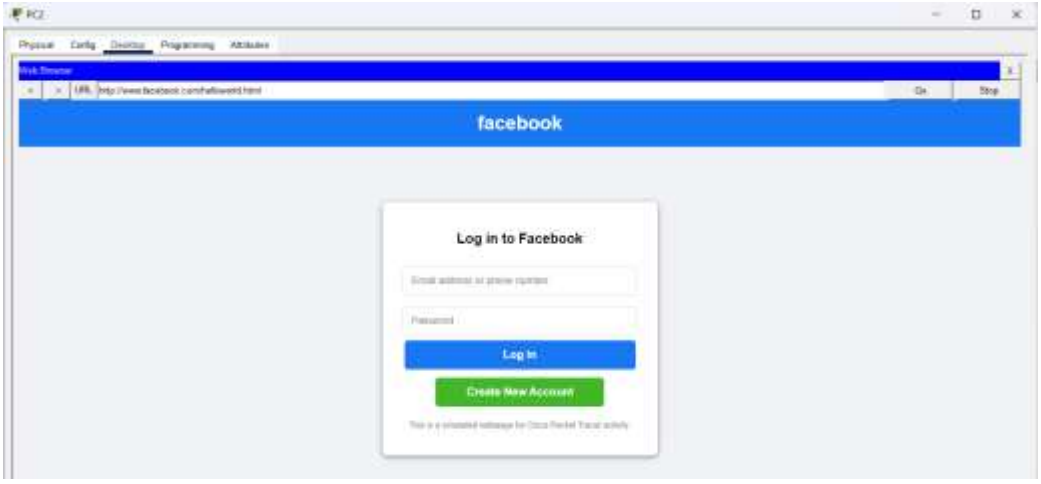

PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.12
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
192.168.1.1

Accessing the web